

Quasi-normal Mode Analysis of Black Hole Ringdown

Joey Wee. MPhil Data Intensive Science, University of Cambridge. Word count: 2908

I. INTRODUCTION

GRAVITATIONAL wave astronomy takes advantage of the propagating distortions in space-time generated by the relative motions of masses to infer the properties of those masses. Advances in the ability to detect these gravitational waves opens up an exciting avenue for the observation of our universe. [1]

Before the epoch of reionisation, the universe consisted of dense hydrogen gas, which scattered photons. As such, it is impossible to make measurements before this period by electromagnetic means. Gravitational wave astronomy yields an opportunity to see past this barrier for the first time.

Binary black hole mergers are among the sources currently accessible to detectors. These are an ideal environment in which to test theories of gravity in a regime inaccessible in Earth's local environment. As such, this coursework will focus on the application of gravitational wave astronomy to the detection and quantification of a black hole merger and using that signal to conduct a consistency test on the predictions of general relativity.

II. THEORY

The *no-hair theorem* states that stationary black holes in a vacuum can be characterised by just their mass and angular momentum. Working in $G = c = 1$ natural units, the angular momentum J is typically written in terms of a dimensionless spin parameter $\chi \in [0, 1]$.

Kerr black holes are rotating black holes to which no-hair theorem applies. These are the resulting objects from the merger of two black holes and are characterised by the Kerr metric. One such merger event will be used to characterise the resulting Kerr black hole in this work.

The black hole merger event has three distinct portions: inspiral, merger, and ringdown. The *inspiral* is when the black holes spiral into one another; this is characterised by a chirp, starting at low amplitude and low frequency then growing to high amplitude and high frequency. Then the *merger* is the actual collision, which is difficult to characterise as the fields are so strong they fail to be described well by a linear approximation. Finally, the *ringdown* which is characterised by a linear perturbation to the Kerr black hole solution. As Kerr black holes are stable, this perturbation oscillates and settles down to the stable solution.

The general ringdown solution is a sum of oscillating decaying functions, the quasi-normal modes (QNMs). QNMs are parameterised by complex frequencies, with the real portion corresponding to the oscillation frequency and the imaginary

portion corresponding to the amplitude decay rate. The QNM solutions are a discrete spectrum labelled by l , m , and n . These indices will be denoted by α . As such, the ringdown signal in the time domain is given by

$$h(t) = \sum_{\alpha} \cos(2\pi f_{\alpha}(t - t_0) + \phi_{\alpha}) \exp\left(\frac{-(t - t_0)}{\tau_{\alpha}}\right),$$

where t_0 is the ringdown start time.

Theory expects the $(m, l, n) = (2, 2, 0)$ mode to be the strongest and the $(3, 3, 0)$ mode to be the second. So, the focus of this work will be on these modes:

$$\alpha \in \{(2, 2, 0), (3, 3, 0)\}.$$

Observations of gravitational waves are made with gravitational wave interferometers. As the waves pass through a region, one of the orthogonal modes to the direction of travel will be compressed and the other expanded, creating a measurable difference in phase between the two arms.

III. QNM MODES (Q1)

The dimensionless complex frequencies $\hat{\omega}$ of the QNMs being examined were extracted from the qnm package.[2] These are related to the complex frequencies by

$$\omega(M, \chi) = \frac{\hat{\omega}(\chi)}{M},$$

where M is the mass of the black hole.

$\hat{\omega}$ for both modes was evaluated over a dense grid of χ and plotted in Fig.1. As χ increases, the rate of damping decreases and the frequency increases, leading to longer-lived, higher-frequency perturbations.

IV. ON AND OFF-SIGNAL DATA (Q2)

It will be useful to perform an analysis on the background noise. To do so, regions of the data with either pure noise or noise and merger signals will have to be identified: the off and on-signal portions respectively. The on-signal portion will later be used to fit for the merger properties.

This is a loose bound as the statistics of the noise will be accurate as long as the off-signal portion clearly does not include any part of the merger signal. Contrastingly, including some off-signal data in the on-signal portion will not affect the ringdown analysis significantly as the QNMs naturally decay to the same mean as the noise.

To identify where the on-signal data started and ended, a simple spectrogram of the data was plotted in Fig.2 to identify the times when the spectral signature deviates from that of the background noise.

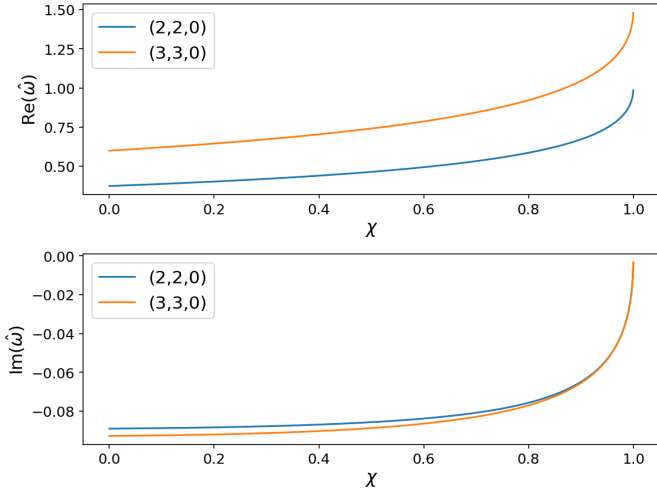


Fig. 1: The real and imaginary frequencies of the two modes plotted against spin. As spin increases, the frequency increases and the damping decreases for both modes. The (3, 3, 0) mode has higher real frequency than the (2, 2, 0) mode across all χ . For low χ , the (2, 2, 0) most is more heavily damped but the values converge at higher χ .

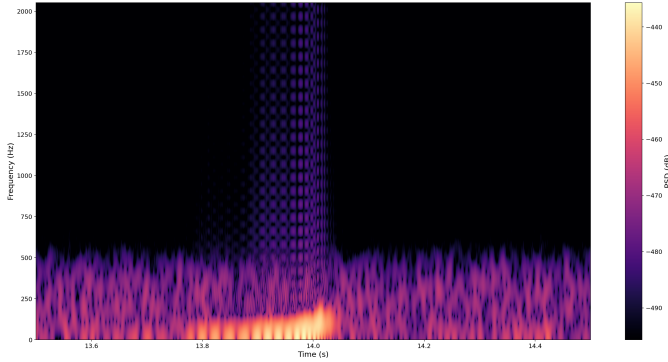


Fig. 2: A spectrogram of the section of data around the signal period. The section of data between 13.8 and 14.0 seconds is dissimilar from the background noise, implying the presence of the merger signal.

The spectrogram shows that the signal has negligible power above 500 Hz for most of the data. However, at around 13.8 seconds, higher frequencies start having power. Similarly, observing the low frequencies, the power significantly increases around 13.8 seconds. Both these effects terminate just after 14 seconds. To ensure that the off-signal statistics are not contaminated, a slightly wider on-signal region between 13.7 and 14.1 seconds is chosen. It can be seen in the plot of strain in Fig.3 that there is no visible deviation away from the noise outside these bounds.

V. RINGDOWN START TIME (Q2)

A suitable t_0 is essential to ringdown analysis of a black hole. If t_0 is chosen too early, the QNM summation model is not applicable, damaging the data fit. However, if t_0 is chosen too late, the fit will be similarly degraded as the QNM signal will have decayed, decreasing the signal-to-noise ratio.

As the QNMs are decaying, it is logical that at any time after the start time t_0 the signal will be weaker than at t_0 . So, a starting estimate of t_0 will be time of the peak amplitude of the strain. A Hilbert transform of the data was used to calculate the instantaneous signal envelope across the data, allowing for a better amplitude estimate than by taking a naïve absolute value of the strain. The peak of this Hilbert envelope was taken as the reference t_0 .

As the inclusion of any merger signal would bias the ringdown analysis and having a later t_0 would only decrease the SNR, the prior effect is more serious. As such, a buffer of 25 milliseconds was added to the peak time to ensure that t_0 existed in the latter regime.

VI. SAMPLING LIKELIHOOD (Q3, Q5-7)

A. Estimating sigma (Q3)

Sampling of the free parameters ($A_1, \phi_1, f_1, \tau_1, A_2, \phi_2, f_2, \tau_2$) in the QNM sum equation was performed using a white noise likelihood

$$\log \mathcal{L}_{\text{WN}}(d | \theta) = -\frac{1}{2} \sum_i \frac{(d(t_i) - h(t_i; \theta))^2}{\sigma^2}.$$

The σ in the likelihood is the standard deviation of the noise. This was estimated from the variance in the strain of the off-signal periods of the data to be

$$\sigma = 8.23 \times 10^{-23}.$$

B. Noise PSD (Q5)

The likelihood operates on the assumption that the underlying noise really is white noise. We can consider how appropriate this assumption is by examining the off-source data's power-spectral density. The PSD was calculated by applying Welch periodogram analyses separately to the sections of the off-source data before and after the on-source period. This was preferred over applying a single periodogram analysis to a simple concatenation of the two off-source periods as it was thought that this would lead to a discontinuity at the boundary, affecting the PSD results by making it appear that subsequent points were less correlated and closer to white noise. These periodograms are plotted in Fig.4.

The periodograms calculated from both portions of the off-source data line up well, implying that the on-off source boundaries were chosen correctly and that neither side contains on-source data contamination. The agreement justifies using the mean of the two PSDs for all further analysis.

The mean periodogram has a peak at the low frequencies which becomes constant at 100 Hz. The taper off around 400 Hz is expected as the strain data being analysed in the coursework has already been subjected to a 20-400 Hz 4th order Butterworth band pass filter.

White noise looks like a constant power-spectral amplitude. While a Butterworth filter can be expected to correlate neighbouring points slightly, the peak at low frequencies is quite significant. As such, it can be concluded that this is not white noise. This agrees with the expected noise profile of the LIGO gravitational wave interferometers. [3]

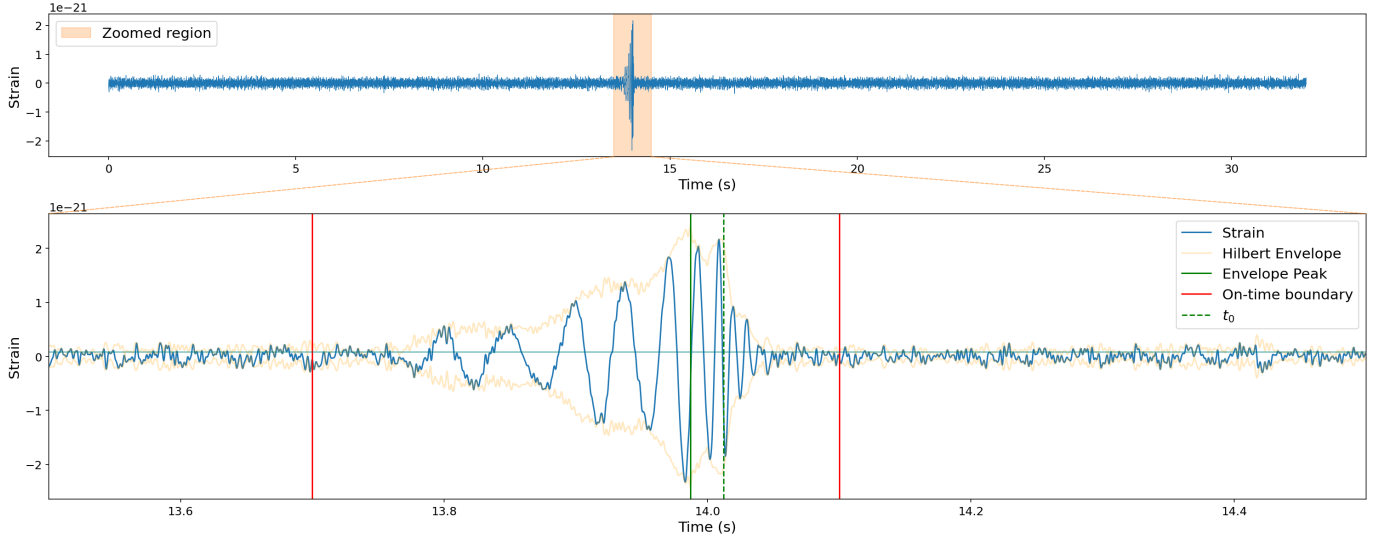


Fig. 3: The time series of the strain around the time of the merger event with the Hilbert amplitude envelope. The times chosen to bound the on-period and ringdown are plotted.

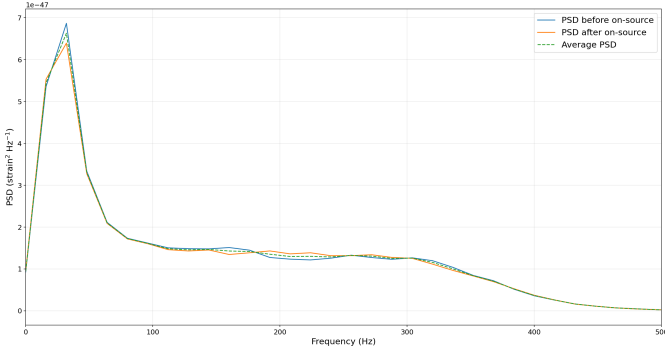


Fig. 4: The power-spectral densities of the noise from both off-periods calculated by Welch periodogram. They line up well so the analysis is performed with the mean thereof.

Fortunately, the QNM signals being studied, when transformed into frequency space, are narrow Lorentzian frequency bands. As such, the signal bandwidth is very narrow compared to the frequency range over which the PSD varies. This means that the noise PSD is approximately flat locally, like white noise. As such, white noise is an acceptable approximation to use.

C. Autocorrelation (Q6)

The suitability of the white noise behaviour can be further examined by calculating the autocorrelation of the noise data at various lags. According to the Wiener-Kinchin theorem, the autocorrelation is the Fourier transform of the PSD. [4] The autocorrelation calculated by Fourier transform is plotted in Fig.5. By considering the σ of the noise and the number of off-source data-points, a noise fluctuation threshold was calculated at 95% confidence and plotted.

A fully white noise signal would yield an autocorrelation with a spike at null lag and no power anywhere else, indicating that the data-points are random and uncorrelated. The

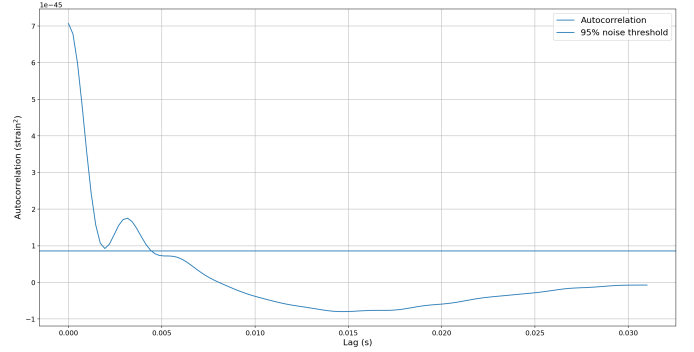


Fig. 5: The autocorrelation of the off-source data and the 95% noise confidence threshold. There is a large peak at low frequencies, similar to what white noise would be expected to look like. There is also a secondary peak well above the 95% noise confidence threshold implying non-whiteness.

autocorrelation calculated shows a large peak at low values, corresponding almost to what white noise would be expected to look like. A portion of the width of the low-lag peak can be attributed to the Butterworth filter correlating nearby points.

However, there is another peak of non-negligible width above the 95% noise fluctuation window. This is likely an artifact from the non-white-noise nature of the off-signal data. Given that the vast majority of the correlation power exists in the very low lags and the magnitude of the secondary peak is much smaller in comparison, the white noise likelihood can still be justified as a reasonable approximation.

D. Autocorrelation sanity check (Q7)

A sanity check can be performed on the correctness of the autocorrelation calculation by the Wiener-Kinchin theorem by comparing the null lag autocorrelation to the off-signal

standard deviation used in the white noise likelihood. The definition of the autocorrelation at lag τ is given by

$$C(\tau) = \langle h(t) \cdot h(t + \tau) \rangle_t.$$

This means that when $\tau = 0$, the $C(\tau)$ is given by the variance of the strain. Comparing the $C(0)$ calculated by the Fourier transform to the σ of the off-signal data, they align to within 0.02%, as expected if the calculation was performed correctly.

VII. SAMPLING (Q4)

The sampling of the posterior distribution of the parameters was achieved using a nested sampler from the dynesty package according to the white-noise likelihood. [5]

The amplitude of the strain data in Fig.3 is of order $\times 10^{-21}$. The specific magnitude of each of the mode magnitudes $A_{1,2}$ is unknown other than that they should be less than the peak strain. As such, a log-uniform prior was used to allow for a range of scales bounded between 1×10^{-25} and 1×10^{-20} .

The phases $\phi_{1,2}$ are bounded between $-\pi$ and π so the prior on the phases was uniform between these two values.

The ranges of the frequencies $f_{1,2}$ to be fit are unknown. It is known that the data has been passed through a bandpass filter with a lower cut-off frequency of 20 Hz so any sample at a lower frequency is unphysical; this was set to be the lower bound of the prior. Similarly, the upper cut-off frequency at 400 Hz was set to be the higher prior bound. As these prior bounds span more than a decade, a log-uniform prior was used.

The data's sampling rate of 4096 Hz corresponds to one reading every 0.24 milliseconds so the lower bound of the decay rate τ was chosen to be 0.1 milliseconds, implying that the decay happens within one datapoint; any smaller decay time would be unphysical. The upper bound is unknown but the literature suggest that τ is typically on the order of milliseconds. [6] So, by setting the prior bound to 100 milliseconds, there is ample room for exploration. Similarly because the prior spans decades, a log-uniform prior was used.

The samples obtained imply good convergence of the sampler as the marginal posteriors of each sampled parameter form Gaussians: the samples are in a likelihood-dominated regime. Marginalising the ϕ and A nuisance parameters, the posteriors on f and τ are plotted in Fig.6.

From these posterior samples the medians and uncertainties of the parameters are extracted. These are printed in Table I and will be used for conducting the consistency test of GR.

TABLE I: Posterior parameter estimates.

Parameter	Value
f_1 (Hz)	$(1.46^{+0.0165}_{-0.0181}) \times 10^2$
τ_1 (s)	$(1.37^{+0.204}_{-0.172}) \times 10^{-2}$
f_2 (Hz)	$(9.61^{+0.00385}_{-0.00403}) \times 10^1$
τ_2 (s)	$(1.34^{+0.0345}_{-0.0341}) \times 10^{-2}$

These values were further verified by plotting the posterior averaged fits and residuals in Fig.7. The posterior averaged fits match the truth very well. While the magnitude of the residuals grows at later times, this is aleatoric uncertainty resulting from the decreasing SNR in ringdown waveforms.

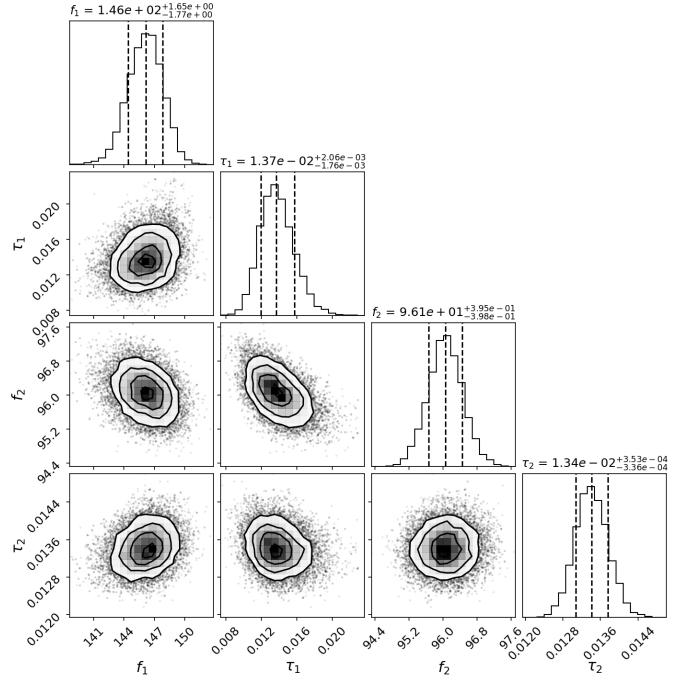


Fig. 6: The posteriors of $f_{1,2}$ and $\tau_{1,2}$. They are all well-constrained, with Gaussian marginals.

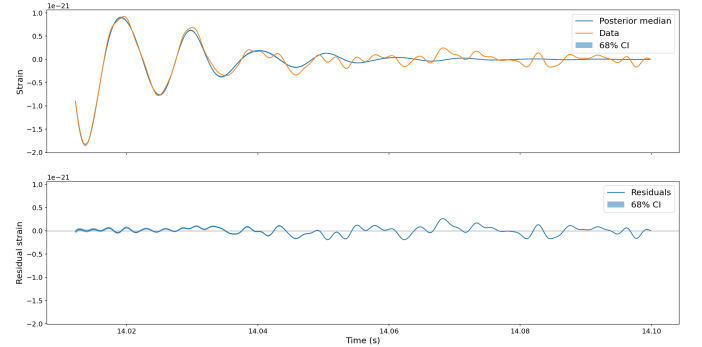


Fig. 7: The top plot shows the posterior averaged fits sampled alongside the truth. The bottom plot plots shows the residuals thereof; these residuals grow with time which is expected as the ringdown signal fades into aleatoric noise.

VIII. CONSISTENCY TESTS (Q8-9)

A. Motivating consistency tests (Q9)

Our everyday environment is characterised by a low velocity and weak gravitational field regime, in which the equations are approximately linear. As such, observations in this linear regime are commonplace and are found to be consistent with GR. However, these tests do not generalise to all environments present in the universe.

Black hole mergers exist at the extreme opposite regime, with high velocities and extremely strong gravitational attraction, where the equations can no longer be approximated as linear. Performing consistency tests on such environments allows us to check GR in a non-linear regime too. Furthermore, due to the extreme nature of the environment, the effects of

small deviations from our theories will be magnified, making it an ideal way to look for physics beyond GR.

Black hole mergers are also experimentally ideal as these GR consistency checks can be achieved with minimal modelling as the remnant Kerr black hole is well described by linear perturbations to the Kerr metric, which result in the QNMs we are fitting. As such we have a very good baseline against which to check for any deviations from GR behaviour.

There are, however, challenges in performing these tests as it is difficult to choose an accurate t_0 and the SNR of the ringdown period we are using for the checks is weak. Furthermore, they require several QNM modes to be detected. This can be difficult as higher modes are less pronounced: overtone detection is still controversial.

B. Performing a consistency test (Q8)

A consistency test by black hole spectroscopy was performed. If the final mass M_f and final spin χ_f of the remnant object calculated from each of the two detected modes are in agreement with one another, the GR theories which were used to make the calculations can be deemed to be consistent.

The complex frequencies returned by qnm have the form

$$\hat{\omega}_\alpha = 2\pi M f - i \frac{M}{\tau}.$$

These complex frequencies across the full range of χ for the two Kerr modes are plotted in the right-hand plot of Fig.8 in dimensions of $\frac{M_f}{\tau_\alpha}$ and $M_f f_\alpha$. These can be compared to the sampled τ s and f s. The sampled parameters are unconstrained in mass so they manifest as lines on the plane.

The intersections of the measurements with the Kerr modes gives that mode's fit's position in $M_f f_\alpha - \frac{M_f}{\tau_\alpha}$ space. It is expected that the (2,2,0) mode will have a lower f_α than the (3,3,0) mode: in Fig.1 the real part corresponding to frequency is lower for all χ for the (2,2,0) mode; as such, the intersection of the lower sampled f with the (2,2,0) mode was calculated and likewise for the (3,3,0) mode with the higher sampled frequency. The M_f of each intersection can be extracted by dividing the $M_f f_\alpha$ coordinate by the sampled f .

χ_f can also be extracted by where along the Kerr mode curve, which is parameterised by χ , the $f\tau$ line intersects. This was performed by finding the χ value at which

$$\frac{|Im[\hat{\omega}_\alpha]|}{Re[\hat{\omega}_\alpha]} = \frac{1}{2\pi f\tau}$$

is satisfied.

This analysis was performed across the posterior sampled combinations of f and τ . The concomitant spread of the χ_f and M_f are calculated and their 68% and 95% confidence intervals are plotted in the left-hand of Fig.8. The (3,3,0) mode has larger uncertainties. This is consistent with the expectation that it is quieter than the dominant (2,2,0) mode and therefore has lower SNR.

In this case, there is weak agreement from the values of the two sets: the median values of the (2,2,0) mode intersect with the 95% confidence interval of the (3,3,0) mode. This measurement only allows us to state weakly that our measurements are consistent with GR. These tests could be improved

upon with higher SNR measurements to better constrain the overtone fit.

C. Checking the ringdown start time

We can use this analysis pipeline to justify our chosen value of t_0 . It would be expected that each of the sampled parameters and their concomitant values of χ and M should be a stable plateau within the range of the ringdown period. The uncertainty of these values would grow at later t_0 as the SNR gets worse and there would be a sharp change in the values when t_0 is before the ringdown start as merger data is included.

The values of τ and f were observed when performing the full analysis on a range of 50 perturbations to the t_0 distributed over ± 15 ms from the initial t_0 to confirm that it is within that plateau.

It shows that the τ values are very stable throughout and the tested Δt_0 . However, even slightly earlier t_0 values show a large disruption in the f sampling. This verifies that the t_0 value chosen for all the above analysis is sound, balancing a good SNR with an earlier ringdown start time with preventing merger signals for polluting the data.

IX. CONCLUSION

In this coursework, analysis of a black hole merger event was performed by fitting QNMs to the ringdown period. The ringdown period of the data was identified by taking the peak strain time and adding a buffer to prevent data contamination; this choice was verified by performing the analysis at small t_0 perturbations, showing that the results of the analysis were stable for a range around the chosen t_0 .

The sampling was performed using a white noise likelihood. Analysis was performed on the noise in the off-signal portions of the data to verify the suitability of the white noise likelihood. While the noise was found to be non-white, it was decided that, because the QNMs are very narrow in frequency space, they looked locally white and, as such, the white noise likelihood is a suitable approximation.

The analysis was used to perform a consistency test on the predictions of GR. It was found that the measurements provide some weak evidence for GR. These can be improved upon with higher SNR measurements to reduce the uncertainty in the overtone detection.

APPENDIX

LLM use statement: LLMs, specifically Claude, were used to support this coursework. They were used to debug code, improve code efficiency and help with some of the more esoteric Matplotlib functions. THE ANALYSIS IS MY OWN.

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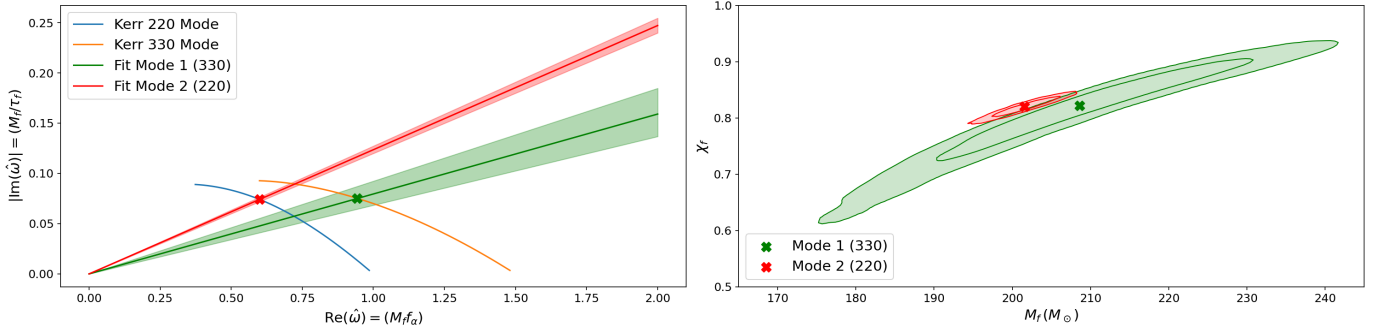


Fig. 8: The calculation of the coordinates of the event in $M_f f_\alpha - \frac{M_f}{f_\alpha}$ space is plotted in the left-hand plot. The concomitant values of χ_f and M_f extracted across the posterior samples are plotted in the right-hand plot.

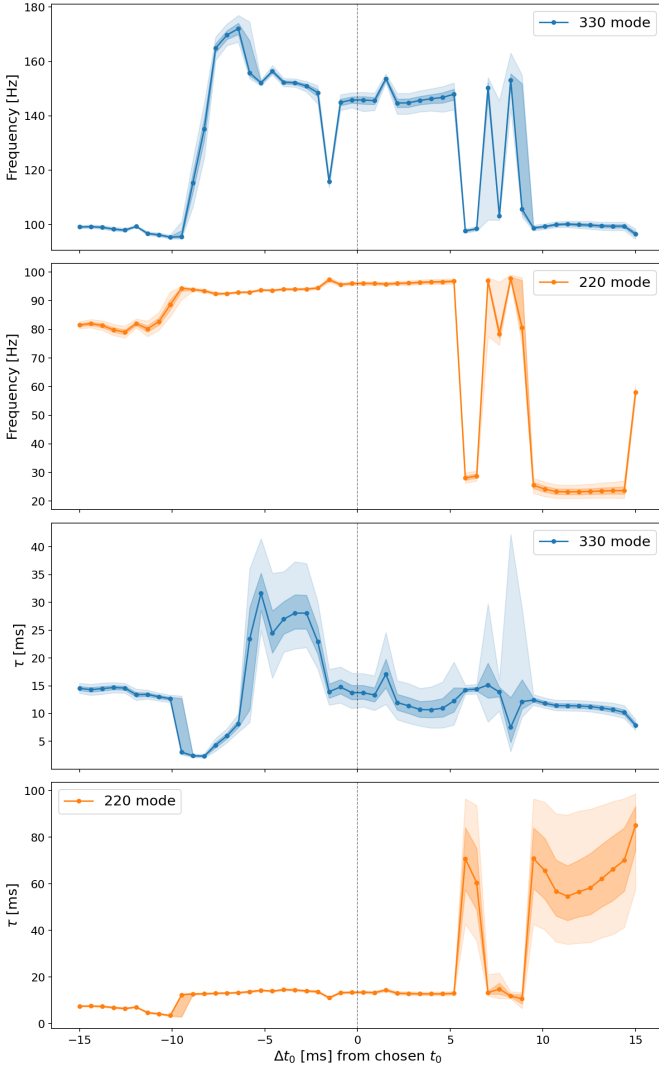


Fig. 9: The plateau analysis run on various perturbed t_0 values. The chosen t_0 fits will within the plateau: earlier values are unstable, justifying the chosen t_0 .

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